

Code-Layout, Readability and Reusability

Date	08 October 2026
Team ID	UNM1611GUFRAN/SWTID-2026-9802
Project Name	AI FAQ Assistant API
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Uniform 4-space indentation used throughout
2	Proper File Structure	Files and folders are logically organized	Yes	Routes, services and models kept in separate folders
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Clear, purpose-based variable names
4	Function / Method Names	Functions are descriptively named	Yes	Descriptive, verb-based function names
5	Code Comments	Inline and block comments explain logic	Yes	Docstrings and inline comments added
6	Modular Design	Code is split into reusable functions/modules	Yes	Logic split into reusable functions and modules
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate and unused code removed
8	Error Handling	Exceptions and errors are handled gracefully	Yes	try/except with proper error responses

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	FAQ Query Handler	Python (FastAPI)	All API endpoints	High

2	Text Preprocessing	Python	Query and FAQ matching	High
3	FAQ Data Loader	Python	Knowledge base updates	Medium
4	Response Generator	Python (AI model)	Chat and FAQ answer routes	High
5	Input Validation Schemas	Python (Pydantic)	All API endpoints	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5 / 5	Clear folder structure with routes, services and models separated
Readability	4 / 5	Consistent naming and indentation make the code easy to follow
Reusability	4 / 5	Core modules can be reused across endpoints and future features
Documentation / Comments	4 / 5	Docstrings and comments added for key functions
Overall Score	4.25 / 5	Well-organized, readable and reusable codebase